Multimodal and Agentic RAG for Grade 10 Mathematics Tutoring in the Nepalese Curriculum

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Abstract

Mathematics education remains challenging for many secondary school students in Nepal, especially in rural areas where access to qualified teachers and structured academic support is limited. This paper proposes a Retrieval-Augmented Generation (RAG)-based math tutoring system for Grade 10 students that provides curriculum-aligned, exam-oriented solutions. The system extends traditional RAG by integrating multimodal capabilities for processing textbook diagrams, agentic workflows for self-reflection and corrective retrieval, and a comparison of specialized math models with general-purpose LLMs. It also addresses a key limitation of text-only RAG systems, which often miss important visual information in mathematical content. The study evaluates different configurations of models, retrieval strategies, and system designs to identify effective approaches for math tutoring.

1 Introduction

1.1 Background and Motivation

Mathematics education in Nepal, particularly at the Grade 10 level, presents significant challenges due to limited access to qualified instruction and structured problem-solving support, especially in rural regions. Students are required to solve examination-oriented problems that demand clear, step-by-step reasoning; however, many rely on memorization rather than developing conceptual understanding. This highlights the need for an accessible and structured learning system that supports curriculum-aligned mathematical problem solving.

Recent advances in large language models (LLMs) have demonstrated strong capabilities in mathematical reasoning and problem solving (Cobbe et al., 2021; Wei et al., 2022). Techniques such as chain-of-thought prompting have been shown to improve reasoning performance in complex tasks (Wei et al., 2022). Despite these improvements, LLMs often produce responses that are not aligned with specific curricula and may lack the structured format required in examinations. Furthermore, they may generate hallucinated or unsupported outputs when relying solely on internal knowledge (Cobbe et al., 2021; Bubeck et al., 2023).

While Retrieval-Augmented Generation (RAG) improves factual grounding by incorporating external knowledge (Lewis et al., 2020), existing systems remain limited in educational settings. They are not aligned with curriculum-specific requirements, cannot effectively handle visual mathematical content such as diagrams, and lack mechanisms for iterative error correction. These limitations reduce their reliability for structured mathematical problem solving.

1.2 Proposed Approach

To address these limitations, we propose an advanced RAG-based mathematics tutoring system that integrates multimodal retrieval and agentic workflows. The system supports both text-based and image-based inputs, where image queries are processed using Optical Character Recognition (OCR) and incorporated into a unified retrieval pipeline.

The multimodal component enables diagram understanding by extracting visual information from textbooks and converting it into structured textual representations. These representations are combined with textual data to support joint text-image retrieval for mathematical problem solving.

In addition, the system incorporates agentic RAG workflows, including Self-RAG, CRAG, and Adaptive RAG, implemented using LangGraph state machines. These workflows allow the system to dynamically decide when to retrieve information, evaluate the quality of retrieved content, and iteratively refine generated solutions, resulting in

improved accuracy and reliability.

1.3 Research Questions

This study is guided by the following research questions:

- **RQ1 (RAG Effectiveness):** Does a curriculum-aligned Retrieval-Augmented Generation system improve the correctness and quality of mathematical solutions compared to baseline LLMs?
- **RQ2 (Multimodal Impact):** How does the inclusion of visual information, such as diagrams, affect the quality and interpretability of mathematical problem-solving?
- **RQ3 (Agentic Workflows):** Do agentic RAG workflows improve solution correctness and reliability compared to static RAG pipelines?

1.4 Method Overview

To answer these questions, we conduct systematic experiments across multiple system configurations, including baseline LLMs, standard RAG pipelines, multimodal RAG, and agentic RAG variants. Performance is evaluated using lexical metrics (BLEU, ROUGE), semantic similarity measures, and efficiency metrics such as response time and latency. This evaluation provides a comprehensive assessment of correctness, structure, and usability.

1.5 Contributions

The main contributions of this work are as follows:

- We design a multimodal and agentic RAG-based math tutoring system tailored for curriculum-aligned learning.
- We conduct systematic experiments comparing multiple RAG variants and model configurations in an educational context.
- We provide insights into the trade-offs between accuracy, cost, and latency for real-world deployment.

The remainder of this paper is organized as follows. Section 2 presents the problem and motivation. Section 3 reviews related work. Section 4 describes the system architecture and methodology. Section 5 presents the evaluation metrics. Section 6 provides experimental results and analysis. Section 7 discusses limitations and implications. Section 8 concludes the study and outlines future work.

2 Related Work

2.1 Large Language Models for Mathematical Reasoning

Recent advances in large language models (LLMs) have demonstrated strong capabilities in mathematical reasoning and problem solving. Prior work shows that models can generate step-by-step solutions when guided by prompting strategies such as chain-of-thought reasoning (Cobbe et al., 2021; Wei et al., 2022). Despite these advances, LLMs often rely solely on internal knowledge, leading to hallucinations, inconsistent reasoning, and outputs that are not aligned with structured curriculum requirements.

2.2 Retrieval-Augmented Generation

Retrieval-Augmented Generation (RAG) improves the factual accuracy of language models by grounding outputs in external knowledge sources (Lewis et al., 2020). Dense retrieval methods further enhance retrieval quality through learned representations (Karpukhin et al., 2020). However, existing RAG research primarily focuses on open-domain question answering, with limited exploration of curriculum-aligned tutoring systems.

2.3 Multimodal Learning

CLIP (Radford et al., 2021) introduced joint text-image representations through contrastive learning, and GPT-4V (OpenAI, 2023) extended these capabilities to vision-language understanding. However, these approaches are largely designed for general image understanding and do not address the structured interpretation of mathematical diagrams.

2.4 Agentic Workflows

Recent work on agentic RAG introduces mechanisms for reflection and retrieval refinement. Self-RAG (Asai et al., 2023) enables models to evaluate their own outputs, while Corrective RAG (CRAG) (Yan et al., 2024) improves retrieval through relevance-based filtering. However, these approaches have primarily been evaluated on general QA tasks rather than structured mathematical

problem solving.

2.5 Research Gap

Existing work on mathematical reasoning with LLMs focuses primarily on improving generation through prompting strategies such as chain-of-thought (Cobbe et al., 2021; Wei et al., 2022). While effective, these approaches rely solely on internal knowledge and lack grounding in curriculum-specific resources, resulting in solutions that are often inconsistent with exam-oriented formats.

Although RAG improves factual grounding (Lewis et al., 2020), current systems are largely designed for open-domain tasks and do not align with structured, step-by-step educational problem solving. Similarly, multimodal models such as CLIP (Radford et al., 2021) and GPT-4V (OpenAI, 2023) are not tailored for interpreting mathematical diagrams, where visual and symbolic reasoning are tightly coupled.

At the same time, agentic RAG methods (Asai et al., 2023; Yan et al., 2024) introduce reflection and retrieval refinement but have not been explored in the context of curriculum-aligned mathematics education.

Therefore, there remains a critical gap in developing a unified system that integrates curriculum-aligned retrieval, multimodal diagram understanding, and agentic reasoning for structured mathematical problem solving.

3 Methodology

3.1 Datasets

The dataset is derived from the official Nepal Grade 10 Mathematics textbook (CDC, 2017). The textbook contains 279 pages organized into 18 chapters, with approximately 200 worked examples, 500 exercises, and more than 150 diagrams. From this source, three main types of data are created: (1) full chapter text, (2) worked examples containing problem statements and step-by-step solutions, and (3) exercise questions. In addition, approximately 150 diagrams are extracted using PyMuPDF and converted into structured descriptions using GPT-4V.

Each example is associated with metadata such as chapter number, chapter title, page number, and example index. For multimodal experiments, each example also includes an `associated_images` field linking it to relevant diagrams. The dataset is stored in JSON format and indexed in ChromaDB for efficient retrieval. This dataset is particularly suitable for the study because it is curriculum-aligned and reflects the exam-oriented mathematical problem-solving context encountered by Grade 10 students in Nepal.

3.2 Data Preprocessing

The preprocessing pipeline consists of five stages for multimodal extraction, as illustrated in Figure 1. This pipeline is designed to transform raw textbook content into structured multimodal representations that can support retrieval and generation.

Stage 1 – Image Extraction: Using PyMuPDF, all embedded images are extracted together with metadata such as page number, position, and dimensions.

Stage 2 – Image Description via GPT-4V: Each extracted image is processed with a structured prompt to generate descriptions of diagram type, labels, measurements, and mathematical concepts.

Stage 3 – Image-Example Linking: Diagrams are associated with relevant worked examples using page proximity and semantic similarity.

Stage 4 – Multimodal Embeddings: Joint representations are created by combining text embeddings from Sentence-BERT (768d), image embeddings from CLIP (512d), and description embeddings (768d), producing 2048-dimensional multimodal vectors indexed in ChromaDB.

Stage 5 – Test Set Creation: A test set of 100 questions is created using stratified sampling, including 33 easy, 34 medium, and 33 hard questions, as well as 40 computational, 35 conceptual, and 25 proof-based questions. Each question is additionally labeled with `visual_required`: true/false.

3.3 System Architecture

The proposed system follows a Retrieval-Augmented Generation (RAG) architecture consisting of four main components designed to support multimodal input processing, adaptive retrieval, and structured solution generation.

(1) Input Processing Layer: This layer handles both text-based and image-based queries. Text queries are processed directly, while image queries are first converted into textual representations using Optical Character Recognition (OCR).

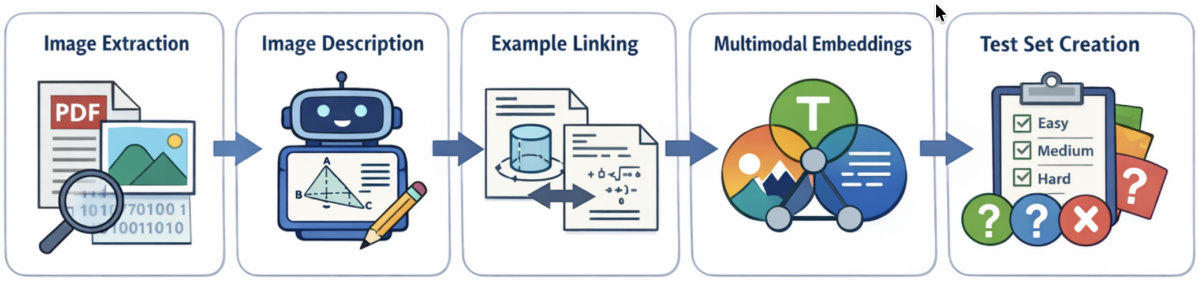

Figure 1: Multimodal data preprocessing pipeline consisting of image extraction, description, linking, embedding, and test set creation.

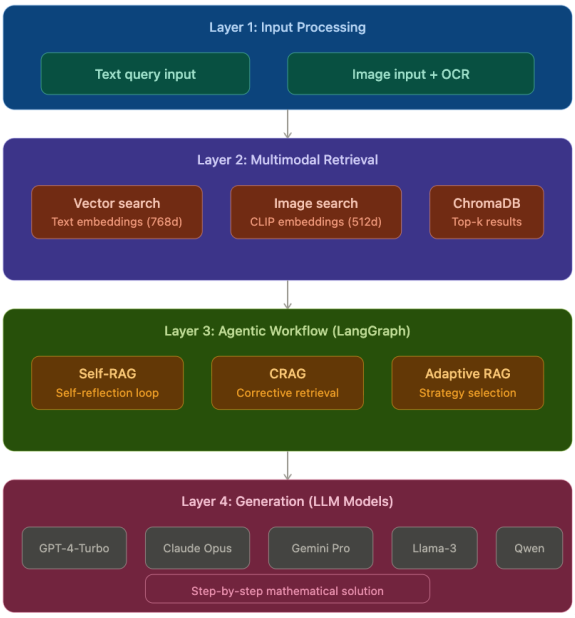

Figure 2: Overall system architecture of the proposed multimodal RAG-based math tutoring system.

This enables a unified input format for downstream retrieval and reasoning.

(2) Multimodal Retrieval Layer: This layer performs joint text-image retrieval over the vector database using ChromaDB. It retrieves the top-k relevant worked examples along with their associated diagrams by leveraging multimodal embeddings. This allows the system to incorporate both textual context and visual information into the reasoning process.

(3) Agentic Workflow Layer: This layer implements adaptive reasoning using LangGraph-based state machines. It supports multiple agentic workflows, including Self-RAG for iterative self-correction, CRAG for retrieval refinement, and Adaptive RAG for dynamic strategy selection. These workflows enable the system to evaluate intermediate outputs and improve solution quality through iterative reasoning.

(4) Generation Layer: This layer generates step-by-step mathematical solutions using multiple large language models. The models used in this study include GPT-4-Turbo, GPT-3.5-Turbo, Claude Opus 4.6, Gemini 1.5 Pro, and Qwen2-Math-7B. Each model is evaluated under both baseline and RAG configurations to compare general-purpose and specialized mathematical reasoning capabilities.

4 Experiments

We design a set of experiments to systematically evaluate the effectiveness of Retrieval-Augmented Generation (RAG), multimodal learning, and agentic workflows for curriculum-aligned mathematical problem solving. The experiments are organized into three groups: (1) baseline evaluation, (2) multimodal RAG, and (3) agentic workflow evaluation.

4.1 Baseline LLM vs RAG

This experiment evaluates whether Retrieval-Augmented Generation (RAG) improves mathematical problem-solving compared to baseline large language models. In the baseline setting, models generate responses using only internal knowledge, whereas in the RAG setting, models are provided with retrieved context from the Grade 10 mathematics textbook.

As shown in Table 1, we compare multiple models under baseline and RAG settings.

4.2 Retrieval Strategy Comparison

This experiment evaluates the impact of retrieval strategies on RAG performance. We compare vector-based retrieval with hybrid retrieval combining semantic and keyword-based matching.

Model	Setting	BLEU	ROUGE-L	BERT
GPT-4-Turbo	Baseline	-	-	-
GPT-4-Turbo	RAG	-	-	-
GPT-3.5-Turbo	Baseline	-	-	-
GPT-3.5-Turbo	RAG	-	-	-
Claude Opus 4.6	Baseline	-	-	-
Claude Opus 4.6	RAG	-	-	-
Qwen2-Math-7B	Baseline	-	-	-
Qwen2-Math-7B	RAG	-	-	-
Gemini 1.5 Pro	Baseline	-	-	-
Gemini 1.5 Pro	RAG	-	-	-

Table 1: Baseline LLM vs RAG comparison.

Model	Modality	BLEU	ROUGE-L	BERT	Time
GPT-4-Turbo	Text-only	-	-	-	-
GPT-4-Turbo	Text+Desc	-	-	-	-
GPT-4V	Text+Image	-	-	-	-
Claude Opus	Text+Image	-	-	-	-
Gemini Vision	Text+Image	-	-	-	-
Qwen2-Math	Text-only	-	-	-	-

Table 2: Multimodal RAG comparison.

4.3 Prompt Style Comparison

This experiment investigates how prompt design affects the quality of generated solutions. We compare general, step-by-step, and exam-oriented prompts.

4.4 Multimodal RAG Experiments

This experiment evaluates the impact of incorporating visual diagram information into the RAG pipeline. We compare text-only retrieval, text with image descriptions, and vision-language models.

As shown in Table 2, we evaluate different multimodal configurations.

4.5 Agentic RAG Experiments

This experiment evaluates agentic RAG workflows such as Self-RAG and CRAG. These approaches introduce reflection and retrieval correction to improve solution quality.

As shown in Table 3, we compare different agentic architectures.

5 Expected Findings

We expect that incorporating Retrieval-Augmented Generation (RAG) will significantly improve the correctness and structure of mathematical solutions compared to baseline language models, due to the inclusion of curriculum-aligned context.

For retrieval strategies, we anticipate that hybrid approaches combining semantic and keyword-based retrieval will provide more relevant context than purely vector-based methods, leading to improved generation quality.

Model	Workflow	BLEU	ROUGE-L	BERT	Time
GPT-4-Turbo	Baseline	-	-	-	-
GPT-4-Turbo	Self-RAG	-	-	-	-
Claude Opus	Self-RAG	-	-	-	-
GPT-4-Turbo	CRAG	-	-	-	-
Claude Opus	CRAG	-	-	-	-

Table 3: Agentic RAG comparison.

In the multimodal setting, we expect that incorporating visual diagram information will improve performance on geometry and visually grounded problems. However, this improvement may come with increased computational cost and latency. We also expect that multimodal approaches will have limited impact on non-visual problem types such as algebra and arithmetic.

For agentic workflows, we hypothesize that Self-RAG and CRAG will improve correctness by enabling iterative refinement and retrieval correction. These approaches are expected to reduce errors caused by irrelevant context, although they may introduce additional latency due to multiple reasoning steps.

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